

Probability, Conditional Probability and Bayes' Theorem

A simple guide with practical banking examples

Concept	Main question
Probability	How likely is an event?
Conditional probability	How likely is an event when another fact is known?
Bayes' theorem	How should probability be updated after receiving new evidence?

1. Probability

Probability describes **how likely an event is to occur**. Its value ranges from 0 to 1, or from 0% to 100%.

$$P(A) = \text{Number of favourable outcomes} / \text{Total number of possible outcomes}$$

Probability	Meaning
0	Impossible
0.5	50% chance
1	Certain

Banking example

A bank has 1,000 transactions, of which 50 are fraudulent.

$$P(\text{Fraud}) = 50 / 1,000 = 0.05 = 5\%$$

Therefore, a randomly selected transaction has a **5% probability of being fraudulent**.

2. Conditional Probability

Conditional probability measures the likelihood of an event **when another event is already known to have occurred**.

$$P(A | B) = P(A \text{ and } B) / P(B)$$

The symbol **P(A | B)** means: the probability of event A, given that event B has occurred.

Banking example

Suppose 200 transactions are international, and 30 of those international transactions are fraudulent.

$$P(\text{Fraud} | \text{International}) = 30 / 200 = 0.15 = 15\%$$

The overall fraud probability was 5%, but it becomes **15%** when we know the transaction is international. Additional information changes the probability.

An important distinction

P(A | B) is generally not equal to P(B | A).

Expression	Question answered
P(Fraud International)	Among international transactions, how many are fraudulent?
P(International Fraud)	Among fraudulent transactions, how many are international?

3. Bayes' Theorem

Bayes' theorem helps us **update an existing probability when new evidence becomes available.**

$$P(A | B) = [P(B | A) \times P(A)] / P(B)$$

Term	Meaning
P(A)	Prior probability: belief before seeing the evidence
P(B A)	Likelihood: probability of the evidence when A is true
P(B)	Overall probability of observing the evidence
P(A B)	Posterior probability: updated belief after seeing the evidence

Fraud-detection example

Assume that 1% of transactions are fraudulent. A fraud-detection system alerts on 90% of fraudulent transactions, but it also alerts on 5% of legitimate transactions.

Given information	Value
P(Fraud)	0.01
P(Alert Fraud)	0.90
P(Alert Not Fraud)	0.05
P(Not Fraud)	0.99

Step 1: Calculate the overall probability of an alert

$$P(\text{Alert}) = [0.90 \times 0.01] + [0.05 \times 0.99] = 0.009 + 0.0495 = 0.0585$$

Step 2: Apply Bayes' theorem

$$P(\text{Fraud} | \text{Alert}) = [0.90 \times 0.01] / 0.0585 = 0.1538 = 15.38\%$$

When an alert is raised, the probability that the transaction is actually fraudulent is approximately **15.38%**. This is not 90% because fraud is rare and the system also produces false alerts for legitimate transactions.

Final recap

Concept	Banking question	Example result
Probability	How likely is fraud?	5%
Conditional probability	How likely is fraud, given an international transaction?	15%
Bayes' theorem	After an alert, how likely is actual fraud?	15.38%